Antiderivatives and Indefinite Integrals

Basic antiderivatives

1 Find each indefinite integral:

a $\int (3x^2 - 4x + 5)\,dx$

b $\int \frac{1}{x^2}\,dx$

c $\int \sqrt{x}\,dx$

d $\int \sec^2 x\,dx$

2 Find $\int (4\sin x + e^x)\,dx$.

Using initial conditions

3 Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.

4 A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.

5 An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s^2.

a Find the velocity function $v(t)$.

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places.